Practical No. : 4

Aim –

Use JavaScript function types, scope, closures; apply try-catch; build a palindrome checker

Software/Tools Required –

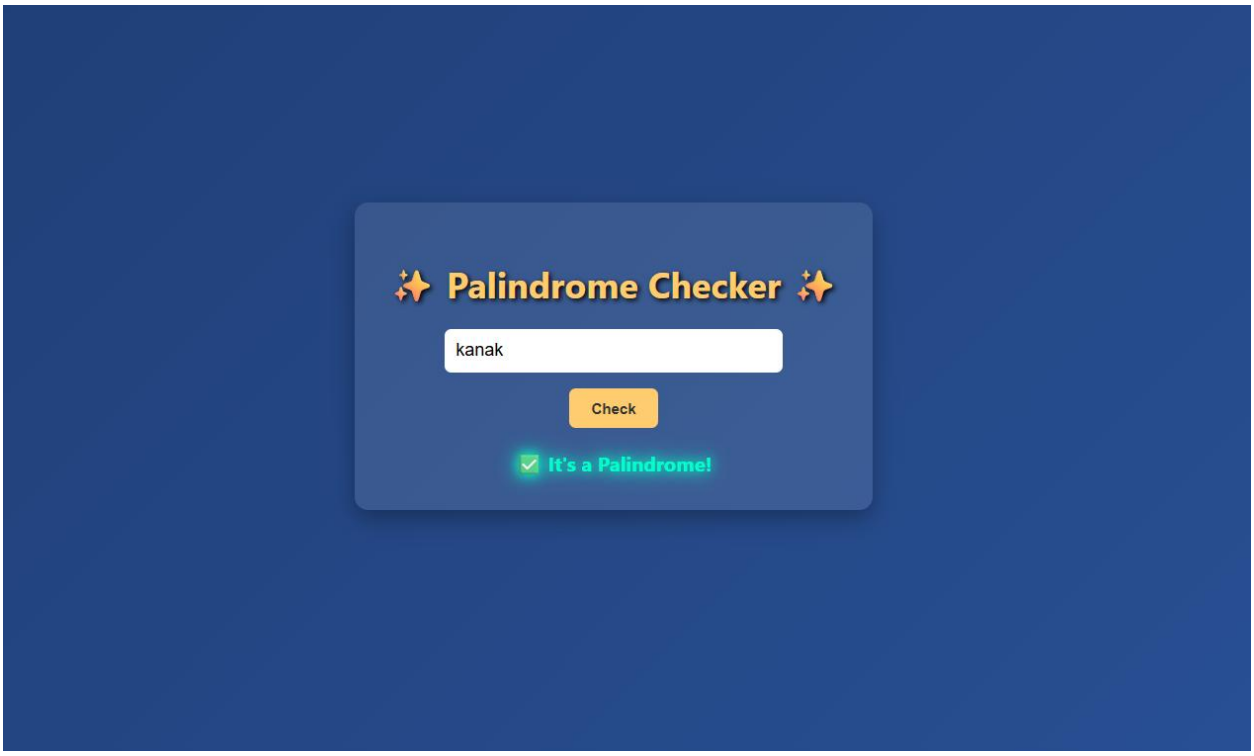
VSCode, Chrome, HTML5, CSS3, Javascript (ES6)

Experiment Program Code –

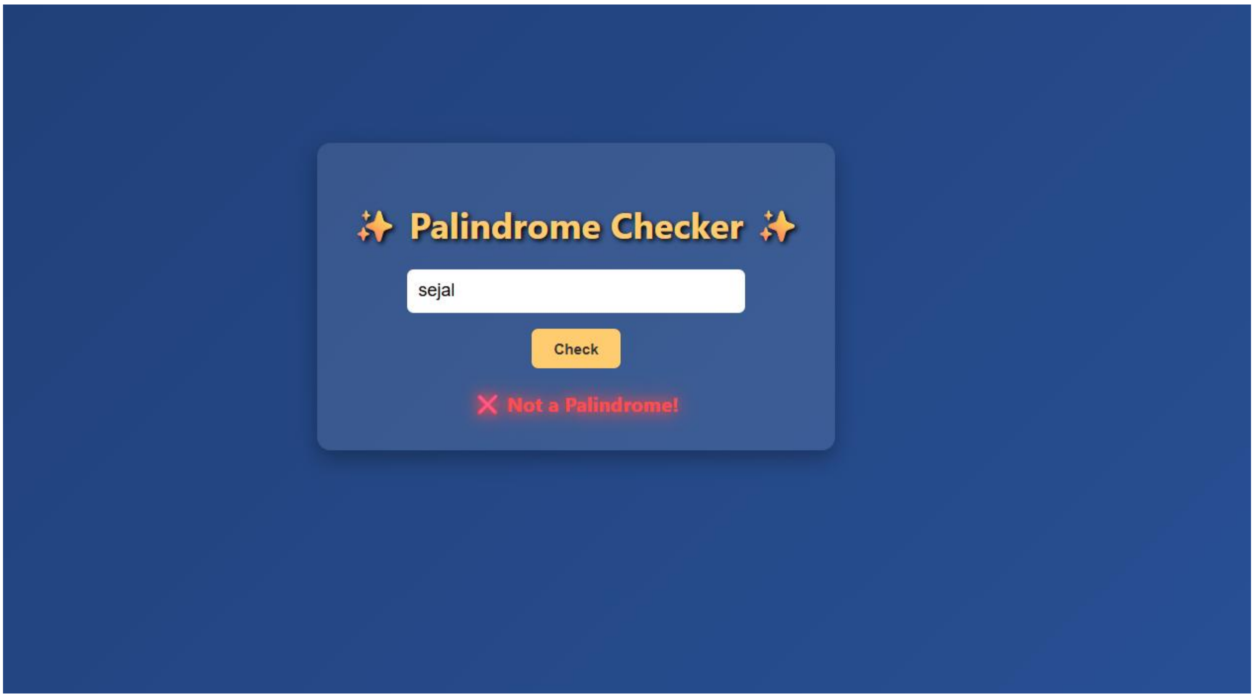
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Palindrome Checker</title>
  <style>
    body {
      font-family: 'Segoe UI', sans-serif;
      background: linear-gradient(135deg, #1e3c72, #2a5298);
      color: #fff;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      margin: 0;
    }
    .container {
      background: rgba(255, 255, 255, 0.1);
      padding: 30px;
      border-radius: 12px;
      text-align: center;
      box-shadow: 0 8px 20px rgba(0,0,0,0.3);
    }
    h1 {
      margin-bottom: 20px;
      color: #ffcc70;
      text-shadow: 2px 2px 5px #000;
    }
    input {
      padding: 10px;
      border: none;
      border-radius: 6px;
      width: 70%;
      font-size: 16px;
```

```
      margin-bottom: 15px;
    }
    button {
      padding: 10px 20px;
      border: none;
      border-radius: 6px;
      background: #ffcc70;
      color: #333;
      font-weight: bold;
      cursor: pointer;
      transition: 0.3s;
    }
    button:hover {
      background: #ffa500;
      transform: scale(1.05);
    }
    .result {
      margin-top: 20px;
      font-size: 18px;
      font-weight: bold;
    }
    .palindrome {
      color: #00ffcc;
      text-shadow: 0 0 10px #00ffcc, 0 0 20px #00ffcc;
    }
    .not-palindrome {
      color: #ff4d4d;
      text-shadow: 0 0 10px #ff4d4d, 0 0 20px #ff4d4d;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>✨ Palindrome Checker ✨</h1>
    <input type="text" id="textInput" placeholder="Enter a word or phrase">
    <br>
    <button onclick="checkPalindrome()">Check</button>
    <div id="result" class="result"></div>
  </div>

  <script>
    function checkPalindrome() {
      const text = document.getElementById("textInput").value.toLowerCase().replace(/[^a-z0-9]/g, '');
      const reversed = text.split('').reverse().join('');
      const resultDiv = document.getElementById("result");

      if (text === "") {
        resultDiv.innerHTML = "⚠️ Please enter some text!";
        resultDiv.className = "result not-palindrome";
        return;
      }

      if (text === reversed) {
        resultDiv.innerHTML = "✅ It's a Palindrome!";
        resultDiv.className = "result palindrome";
      } else {
        resultDiv.innerHTML = "❌ Not a Palindrome!";
        resultDiv.className = "result not-palindrome";
      }
    }
  </script>
</body>
</html>
```

Output –



✨ Palindrome Checker ✨
kanak
Check
✅ It's a Palindrome!



✨ Palindrome Checker ✨
sejal
Check
❌ Not a Palindrome!

Case Study Title –

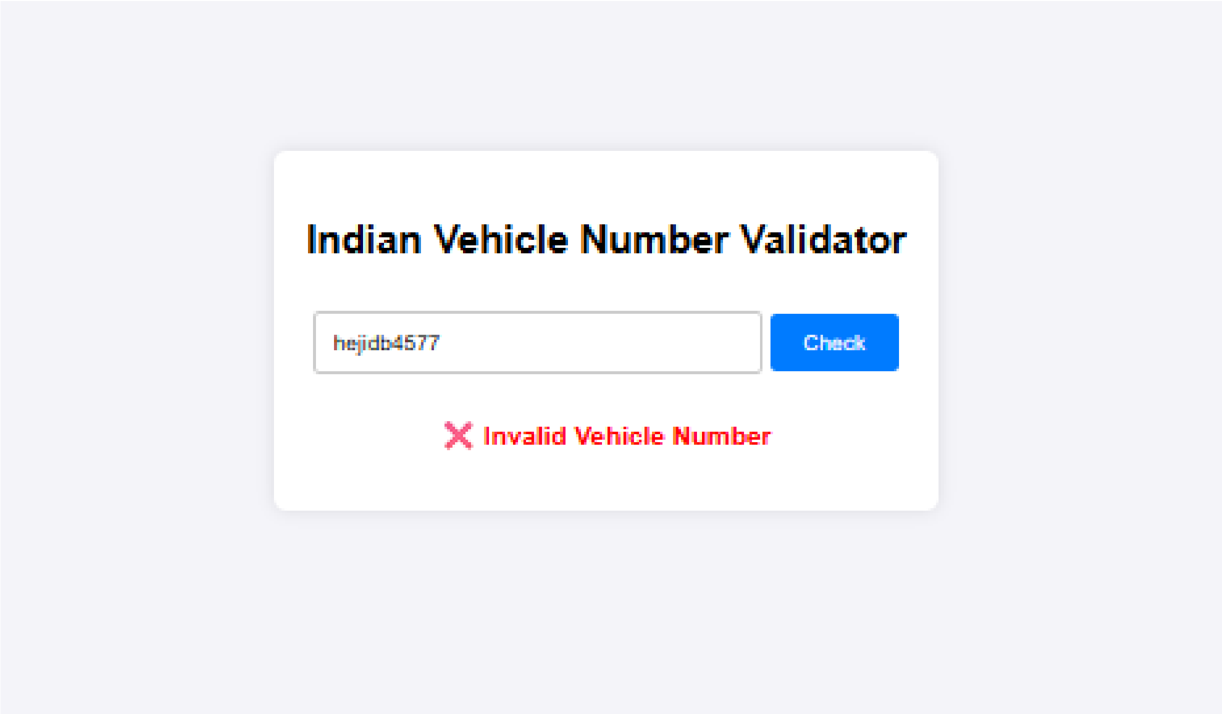
To develop a web application using HTML and JavaScript that validates a vehicle registration number using JavaScript functions and exception handling (try-catch) based on predefined validation rules, and displays whether the registration number is valid or invalid.

Case Study Program Code –

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Indian Vehicle Number Validator</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f4f4f9;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }
    .container {
      background: white;
      padding: 20px;
      border-radius: 8px;
      box-shadow: 0 0 10px rgba(0,0,0,0.1);
      text-align: center;
    }
    input {
      padding: 10px;
      width: 250px;
      margin: 10px 0;
      border: 1px solid #ccc;
      border-radius: 4px;
    }
    button {
      padding: 10px 20px;
      background: #007bff;
      color: white;
      border: none;
      border-radius: 4px;
      cursor: pointer;
    }
    button:hover {
      background: #0056b3;
    }
```

```
    #result {
      margin-top: 15px;
      font-weight: bold;
    }
  </style>
</head>
<body>
  <div class="container">
    <h2>Indian Vehicle Number Validator</h2>
    <input type="text" id="vehicleNumber" placeholder="Enter vehicle number (e.g.,
MH12AB1234)">
    <button onclick="validateNumber()">Check</button>
    <p id="result"></p>
  </div>

  <script>
    function validateNumber() {
      const number = document.getElementById("vehicleNumber").value.trim();
      const result = document.getElementById("result");

      // Regex for Indian vehicle number: AA00AA0000
      const regex = /^[A-Z]{2}[0-9]{2}[A-Z]{1,2}[0-9]{4}$/;

      if (regex.test(number)) {
        result.textContent = "✅ Valid Vehicle Number";
        result.style.color = "green";
      } else {
        result.textContent = "❌ Invalid Vehicle Number";
        result.style.color = "red";
      }
    }
  </script>
</body>
</html>
```

Output –


Indian Vehicle Number Validator
hejidb4577
Check
❌ Invalid Vehicle Number

Conclusion –

The experiment was successfully performed to understand JS functions, variable scope, closures, and exception handling using try-catch. A Vehicle Registration Validator and a Palindrome Checker were developed, these applications demonstrated input validation, string manipulation, and error handling using JavaScript.